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EDUCATION

The University of Chicago - M.S. Applied Data Science
Coursework: Time Series Forecasting, Machine Learning Operations, NLP
MPSTME, NMIMS University - B.Tech. Computer Science
Specialization: Business Systems

Chicago, IL
Sep 2024 - Dec 2025
Mumbai, IN
Jul 2020 - Jul 2024

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, R

AI/LLMs: LangChain, LangGraph, RAG, Agentic AI, QLoRA, vLLM, HuggingFace, PyTorch, TensorFlow, Scikit-learn

Backend/Tools: FastAPI, Rest APIs, MCP, Pandas, Docker, Tableau, Power BI

Cloud & MLOps: AWS (EC2, Sagemaker), GCP (BigQuery,), Azure (Azure ML), Airflow, Spark, MLflow, GitHub Actions

EXPERIENCE

Tolan Center for Healthcare – Chicago Booth

Chicago, IL

Lead Software Engineer

Feb 2026 - Present

- Shipped a multiuser educational platform from a single-server setup to a multi-tenant SaaS platform on AWS (ECS Fargate, RDS, ALB), enabling enterprise licensing and independent deployment for university clients
- Engineered a streaming ETL pipeline on Databricks using Structured Streaming and Delta Lake to ingest EHR clinical events into PostgreSQL, processing 100K+ records daily with sub-4s freshness lag and <0.1% error rate

Chicago Transit Authority

Chicago, IL

Data Science Intern

May 2025 – Dec 2025

- Reduced data load time and lowered infrastructure costs by ~25% by designing and deploying an AWS ETL pipeline migrating on-prem Oracle data to Redshift for scalable cloud data warehousing
- Processed 1.1M+ records daily with reliable execution by building automated ETL workflows using AWS Glue, Lambda, and Airflow with modular DAGs, SQL transformations, and monitoring/failure handling

Inference Analytics AI

Chicago, IL

Artificial Intelligence Engineer

Mar 2025 - Dec 2025

- Engineered a RAG pipeline over 70K+ ICD-10 descriptions using MedCPT embeddings and Milvus, enabling sub-second clinical text retrieval; presented the architecture directly to enterprise healthcare stakeholders
- Designed a two-stage retrieval architecture with hybrid BM25 + vector search and cross-encoder reranking, improving context recall from 0.71 to 0.88 (+23.9%) with semantic caching via AWS ElastiCache (Redis)
- Improved clinical coding accuracy by fine-tuning Llama3-OpenBioLLM-8B with QLoRA on clinical note to ICD-10 mappings, served via vLLM on AWS EC2, achieving 82% end-to-end accuracy and 0.91 RAGAS faithfulness

PROJECTS

Demand Forecasting & A/B Testing | Python, Pandas, Numpy, Keras, Tensorflow

- Scaled weekly demand processing across 59M+ records and 30,490 store-SKU series using Pandas/NumPy, with feature engineering and Tableau dashboards analyzing sparsity, seasonality, and store-level demand trends
- Improved forecast accuracy 41% over baseline (WMAPE 0.62→0.36) by benchmarking time-series (ETS), machine learning (per-store LightGBM), and deep learning (LSTM) models with rolling-origin validation
- Drove a validated 14.1% inventory-cost reduction (n=30,474, p<0.001) through pre-registered offline A/B testing with service-level guardrails, pairing quantile-loss ordering with newsvendor simulation

Multi-Agent Coder | LangGraph, LangChain, Celery, MCP, FastAPI, ChromaDB, Python, TypeScript

- Built a LangGraph-based multi-agent coding system (coding, testing, security, documentation) with graph orchestration, queue-based routing, and core data structures, improving coding benchmarks by 20% over a single-model baseline.
- Designed fault-tolerant LLM infrastructure with circuit breakers, semantic caching, prompt caching, reducing latency and cost across a hybrid local/cloud routing layer
- Implemented an event-sourced async task pipeline with Celery, MCP integrations for browser automation and GitHub, and a vector-based episodic memory layer to improve agent decision-making using past task outcomes

Real-Time Anomaly Detection Pipeline with MLOps | Kafka, Redis, Azure ML, Scikit-learn, PyTorch, MLflow

- Designed a real-time Azure inference pipeline streaming live data through Kafka into a Redis feature store, serving an Isolation Forest + Autoencoder ensemble through a Managed Online Endpoint
- Automated the full ML lifecycle with a 5-stage cloud training pipeline, ROC AUC quality gate blocking underperforming models from production, and weekly retraining with zero-touch redeployment of the best registered model
- Implemented CI/CD using GitHub Actions with automatic component redeployment on every push, OpenTelemetry observability, a live KQL monitoring dashboard, and automated alerts detecting pipeline failures in production

Sports Odds Prediction Pipeline | Python, Spark, GCP BigQuery, Vertex AI, DVC, PyTorch

- Architected an ELT pipeline using Spark and GCP BigQuery with Vertex AI Feature Store for low-latency feature serving, and trained XGBoost and PyTorch models to predict pre-game American betting odds, achieving an MAE of 11.4 points